

Name of the Institution
Name of the Subject
PARTIAL / FINAL EXAM

Student Name: _____

Date: _____ Group: _____

Total Score: _____ Score Obtained: _____

General Instructions

- Read all questions carefully before answering.
- Use blue or black pen. Pencil answers are not accepted.
- The duration of the exam is **90 minutes**.
- The score for each question is indicated at the beginning of the question.
- Clarity, order, and justification of the answers will be valued.

Question 1 (2 points)

Answer the following questions on topic X.

- a) Explain briefly the concept of *Fundamental Theorem of Calculus*.
- b) Mention two practical applications of derivatives in engineering.

Question 2 (3 points)

Solve the following linear algebra problem.

- a) Calculate the inverse of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.
- b) Determine whether the vectors $v_1 = (1, 0, 1)$, $v_2 = (0, 1, 1)$ and $v_3 = (1, 1, 0)$ are linearly independent.

Question 3 (2.5 points)

Prove the following proposition.

- a) Prove that if a function $f(x)$ is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) , then there exists at least one point $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$. (Mean Value Theorem)

Question 4 (2.5 points)

Application problem.

- a) A company produces Q units of a product, where the total cost of production is given by the function $C(Q) = 0.01Q^2 + 5Q + 100$. If the selling price per unit is \$20, determine the quantity of units the company should produce to maximize its profit.

Good luck!